Chair of Mathematical Statistics
Department of Mathematics
Technical University of Munich

E0001
Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Statistics for Business Administration

Exam:	MA9712 / Final-MA9712-22	**Date:**	Thursday 11th August, 2022
Examiner:	Prof. Mathias Drton	**Time:**	17:15 – 18:45

Working instructions

- This exam consists of **18 pages** with a total of **9 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 52 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **scientific pocket calculator**
  - the **cheat sheet**, which is contained in this exam on page **16**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from ___________ to ___________ / Early submission at ___________

☐ Exam empty

Problem 1 (7.5 credits)

Survey USA asked 500 randomly sampled San Franciscans the following question:

"The FBI has a court order demanding Apple help it unlock the iPhone belonging to the San Bernardino shooter. Apple says creating the custom software this would require would set a dangerous precedence and create a back door that, in the wrong hands, could potentially be used to unlock ANY iPhone. Do you think Apple should? Or Should not? Comply with the court order in this particular case?"

The distribution of responses by age group is shown below.

```
##             Age_Group
## Opinion      18-34 35-49 50-64 65+ Sum
##   Not sure      32    12    15  11  70
##   Should        64    55    88  65 272
##   Should not    50    51    35  22 158
##   Sum          146   118   138  98 500
```

0 1 a) In evaluating the relationship between opinion (on whether Apple should comply with the court order or not) and age, what is the response variable and what is the explanatory variable? Explain your reasoning in one sentence.

0 1 b) Suppose a statistically significant relationship is found between opinion and age. To whom can the results be generalized and can a causal relationship be inferred? Explain your reasoning.

0 1 c) Decide if a scatterplot is an appropriate visualization for these data. Justify your answer. If that's not the case, propose an appropriate visualization.

d) Consider the joint probability distribution defined through the above contingency table. Does there appear to be a relationship between opinion and age? Clearly state any probabilities you use as justification.

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e) Consider the joint probability distribution defined through the above contingency table. Show, that the probability of a randomly selected *18-34 year old* thinks Apple *should not* comply with the court order is equal to 0.342 (rounded to three decimal places).

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f) What is the probability that in a random sample of five *18-34 year olds* exactly two believe that Apple *should not* comply with the court order?
Hint:

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```
dbinom(2, size = 5, prob = c(0.292, 0.342, 0.438))
## [1] 0.303 0.333 0.341

pbinom(2, size = 5, prob = c(0.292, 0.342, 0.438))
## [1] 0.847 0.777 0.615
```

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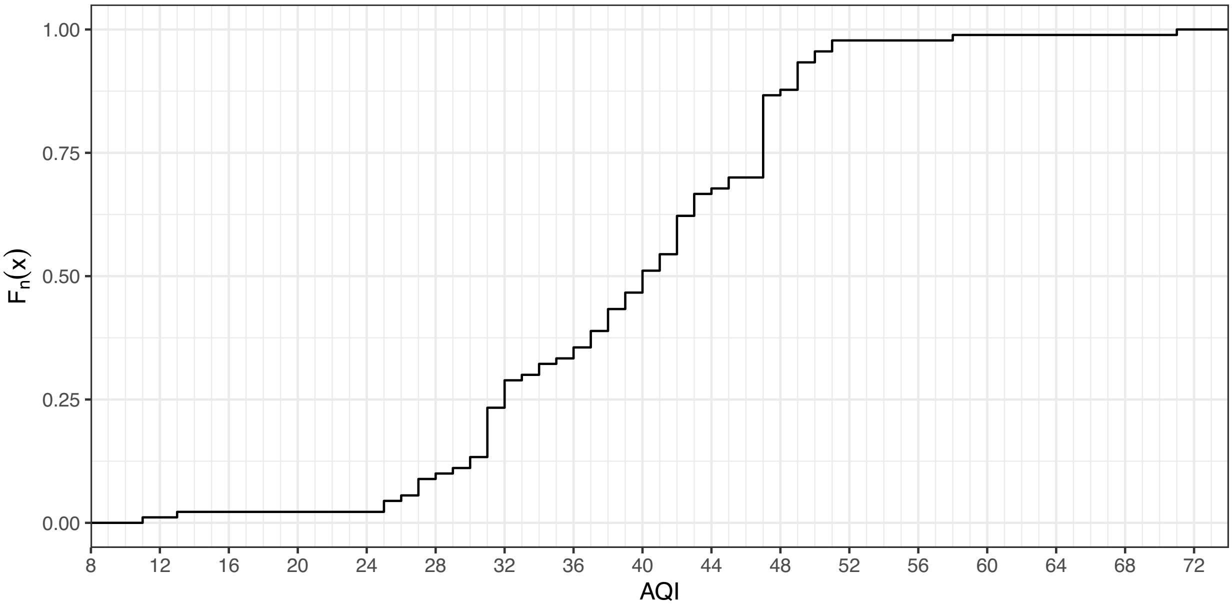
Problem 2 **(6 credits)**

Daily air quality is measured by the air quality index (AQI) reported by the Environmental Protection Agency. This index tells you how clean or polluted your air is, and what associated health effects might be a concern for you. The index is calculated for five major air pollutants regulated by the Clean Air Act, one of which is particulate matter. Values of this index range from 0 to 300 and a higher value indicates lower air quality.

AQI was reported for 90 randomly sampled days in Durham over the past 5 years. The empirical distribution function F_n below shows the distribution of these values. These 90 days represent a reasonably random and representative sample, and you may assume that the AQI levels for these days are independent.



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a) Compute the interquartile range.

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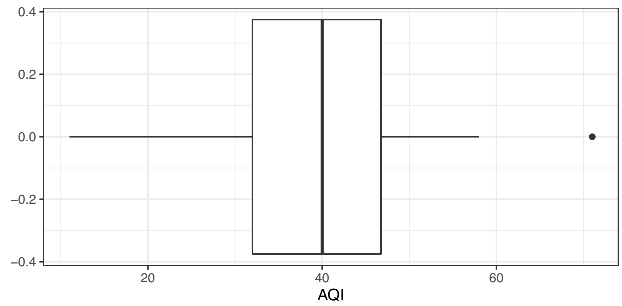
b) We would like to use these data to construct a confidence interval for the average AQI in Durham. Evaluate, in context of the data, whether the conditions for inference based on the central limit theorem are reasonably met.

Hint:



c) The empirical mean and standard deviation of these 90 AQI values is 39.26 and 9.13, respectively. Justify that a two-sided 90% confidence interval for the true average AQI in Durham is equal to $(37.68, 40.84)$ (rounded to two decimal places).
Hint:

```
qnorm(c(0.025, 0.05, 0.1, 0.9, 0.95, 0.975))
## [1] -1.96 -1.64 -1.28  1.28  1.64  1.96
```

0 1 2

d) Interpret the confidence interval from the previous part in the context of the data.

0 1

Problem 3 (4 credits)

Jose uses the parking lot at the train station Olympiazentrum every Thursday evening. However, some days the parking lot is full, often due to events in the nearby Olympiapark. There are cultural events on 33% of evenings, sporting events on 22% of evenings, and no events on 45% of evenings. When there is a cultural event, the parking lot fills up about 21% of the time, and it fills up 75% of evenings with sporting events. On evenings when there are no events, it only fills up about 4% of the time.

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If Jose comes to the parking lot and finds it full, what is the probability that there is a sporting event? Write down all steps, which are needed to compute the wanted probability. In case you use a tree, properly label the nodes in the tree.

Problem 4 (5 credits)

In a February 2016 Public Policy Polling survey, respondents were asked which team they would root for at the next basketball game between Duke University and University of North Carolina (UNC). The responses are summarized below.

```
addmargins(table(duke_unc))

##        Team
## Sex      Duke UNC Sum
##   Man     188 225 413
##   Woman   178 287 465
##   Sum     366 512 878
```

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a) We would like to test if there is a difference between the proportions of men and women who will root for Duke on the next basketball game between these two teams using a randomization test. State the null and alternative hypothesis of this test problem.

b) Show that the observed sample statistic is equal to 0.072 (rounded to three decimal places).

0 1 2

c) The simulated null distribution was computed using the following code

0 1

```
library(infer)
null_dist <- duke_unc %>%
  specify(Team ~ Sex, success = "Duke") %>%
  hypothesize(null = "independence") %>%
  generate(reps = 200) %>%
  calculate(stat = "diff in props", success = "Duke", order = c("Man","Woman"))
```

Calculate the p-value using the following output

```
null_dist %>%
  filter(abs(stat) >= 0.072) %>%
  group_by(stat) %>%
  summarize(n = n())

## # A tibble: 6 x 2
##      stat     n
##     <dbl> <int>
## 1 -0.0922     1
## 2 -0.0830     1
## 3 -0.0785     1
## 4  0.0770     1
## 5  0.0816     1
## 6  0.0861     1
```

d) Using $\alpha = 0.05$, state the conclusion of the hypothesis test. Justify your answer. Interpret the conclusion in context of the data and the hypotheses.

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Problem 5 (4 credits)

Determine if the following statements are **true** or **false**. Justify your answer
Remark: Only correct justifications are awarded with points. Not the decision (true/false).

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a) In order for the sampling distribution of a mean to be nearly normally distributed, the expected numbers of successes and failures must be at least 10.

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b) Random sampling is sufficient for finding evidence of a causal relationship.

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c) Suppose a researcher sets the significance level at 5% prior to conducting an experiment. After conducting the study, the researcher computes a p-value of 0.04. The researcher can be certain that the null hypothesis is false.

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d) If a given value (for example, the null hypothesized value of a parameter) is within a 90% confidence interval for a population parameter, it will definitely also be within a 95% confidence interval for the same poplulation parameter calculated based on the same sample.

Problem 6 (3 credits)

A psychologist wants to determine if socio-economic status is related to game playing preferences. Fifty children, in total, were identified from families of low, middle, and high socio-economic status, and then the children were asked to select one of the games: Monopoly, Battleship, or Connect Four. They performed a chi-squared test which yielded a test statistic value of 5.0.

a) State the null hypothesis H_0 of the chi-squared test.

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b) State a range (p^L, p^U) for the p-value of the chi-squared test, which is as small as possible, using the values in the hint.
Hint:

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```
qchisq(c(0.05, 0.1, 0.2, 0.3), df = 3, lower.tail = FALSE)
## [1] 7.81473 6.25139 4.64163 3.66487

qchisq(c(0.05, 0.1, 0.2, 0.3), df = 4, lower.tail = FALSE)
## [1] 9.48773 7.77944 5.98862 4.87843
```

Problem 7 (9 credits)

In a study that investigates the impact of the Danish Action Plan for the Aquatic Environment, which addresses pollution of the Danish water resources, the concentration of nitrogen (measured in g/m^3) was measured in a particular river in 1998, 2003, and 2011. Six measurements were randomly taken in each year. A summary of the nitrogen concentration is provided.

```
library(tidyverse)
AE %>%
  group_by(year) %>%
  summarize(mean = mean(nitrogen),
            sd = sd(nitrogen),
            n = n()
  )

## # A tibble: 3 x 4
##   year   mean    sd     n
##   <chr> <dbl> <dbl> <int>
## 1 1998   5.55 0.486     6
## 2 2003   5.19 0.362     6
## 3 2011   4.05 0.673     6
```

We want to test at the 5% significance level the relationship between the two variables using an ANOVA model.

0 1

a) State the null and alternative hypotheses for testing the relationship between the two variables?

0 1 2

b) What are the conditions for ANOVA? Evaluate whether each of these conditions are satisfied in the context of these data. **Note any additional information you would need** to fully evaluate the conditions.

c) Replace the missing values (NA) in the following ANOVA table. Write the missing values in the table given in the box. Only the values added in the table will be graded.

0 1 2 3

```
summary(aov(nitrogen~year, data=AE))
            Df Sum Sq Mean Sq F value  Pr(>F)
year        NA   7.36      NA   13.47 0.00045 ***
Residuals   NA     NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hint:

```
var(AE$nitrogen) * 17
## [1] 11.46
```

```
summary(aov(nitrogen~year, data=AE))
            Df              Sum Sq       Mean Sq       F value        Pr(>F)
year                          7.36                       13.47       0.00045 ***

Residuals

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d) State the conclusion of the hypothesis test. Justify your answer. Interpret the conclusion in context of the data and the hypotheses.

0 1

e) What percent of total variability in nitrogen concentration is explained by year?

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f) Assume you want to conduct pairwise t-tests in a post-hoc analysis using a Bonferroni correction. Determine the significance level, which should be used for each of these t-tests.

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Problem 8 (8.5 credits)

The Child Health and Development Studies considered in a study all pregnancies between 1960 and 1967 among women in the Kaiser Foundation Health Plan in the San Francisco East Bay area. Therefore, they collected a random sample of 800 mothers. Variables of interest include `bwt` (birth weight of the child in ounces), `parity` (1 - first born child, 0 - otherwise), `gestation` (length of pregnancy in days), `age` (mother's age in years), `height` (mother's height in inches), `weight` (mother's pregnancy weight in pounds) and `smoke` (1 - mother is a smoker, 0 - otherwise).

```
Call:
lm(formula = bwt ~ ., data = babies)

Residuals:
    Min      1Q  Median      3Q     Max
-54.249 -10.315  -0.135  10.269  51.463

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -108.46339   17.29916  -6.270 5.93e-10 ***
gestation      0.53578    0.03701  14.476  < 2e-16 ***
parity        -3.65823    1.33525  -2.740  0.00629 **
age            0.11144    0.10270   1.085  0.27819
height         1.15701    0.24055   4.810 1.81e-06 ***
weight         0.04087    0.03129   1.306  0.19196
smoke         -8.07276    1.14557  -7.047 3.98e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.72 on 793 degrees of freedom
Multiple R-squared:     NA,Adjusted R-squared:  NA
F-statistic: 54.48 on 6 and 793 DF,  p-value: < 2.2e-16
```

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a) State the equation of the estimated model.

Remark: You can round the estimated coefficients to two decimal places.

0 1 2

b) Interpret the estimated slopes for `gestation` and `smoke`.

c) State the null and alternative hypothesis, which are tested on the line where the p-value is 0.006287.

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d) Calculate the adjusted R^2 value.
Hint:

```
var(babies$bwt)
## [1] 346.472
```

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e) Now we apply the AIC approach to reduce the model with a backward elimination. Use the following output to decide if a variable can be dropped in the first step of the backward elimination. Justify your answer.

Remark: Be aware that the given output is not the standard output of `step()`. The rows have been permuted.

```
step(model)
Start:  AIC=4415.11
bwt ~ gestation + parity + age + height + weight + smoke

             Df Sum of Sq    RSS    AIC
- gestation   1     51803 247828 4600.7
- height      1      5719 201744 4436.1
- weight      1       422 196447 4414.8
<none>                    196026 4415.1
- parity      1      1855 197881 4420.6
- age         1       291 196317 4414.3
- height      1      5719 201744 4436.1
- smoke       1     12276 208301 4461.7
```

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Problem 9 (5 credits)

The RMS Titanic sank in the early morning hours of 15 April 1912 in the North Atlantic Ocean. The dataset `titanic` contains information about passengers aboard of the RMS Titanic

```
titanic
```

```
## # A tibble: 714 x 7
##    Survived Pclass Sex      Age SibSp Parch  Fare
##       <int>  <int> <chr>  <dbl> <int> <int> <dbl>
##  1        0      3 male      22     1     0  7.25
##  2        1      1 female    38     1     0 71.3
##  3        1      3 female    26     0     0  7.92
##  4        1      1 female    35     1     0 53.1
##  5        0      3 male      35     0     0  8.05
##  6        0      1 male      54     0     0 51.9
##  7        0      3 male       2     3     1 21.1
##  8        1      3 female    27     0     2 11.1
##  9        1      2 female    14     1     0 30.1
## 10        1      3 female     4     1     1 16.7
## # ... with 704 more rows
```

We are interested in predicting the survival of the passenger (`Survived` - 1="yes", 0="no") using the variables `Pclass` (Passenger Class), `Sex`, `Age`, `SibSp` (number of sibling/spouses aboard), `Parch` (number of parents/children aboard) and `Fare` as predictor variables.
We split the data into a training set `titantic_train` and a test set `titanic_test`.

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a) Since we have binary response variable, a logistic regression model will be fitted to the data. Add the missing part to the following code, such that a logistic regression model using all variables as predictor variables will be fitted to the training data.

```
model_full <- glm(Survived ~ ., data = titanic_train, ___)
```

```
model_full <- glm(Survived ~ ., data = titanic_train,                    )
```

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b) A summary of the fitted model is the following

```
## # A tibble: 7 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)  4.77      0.674       7.08  1.41e-12
## 2 Pclass      -0.997     0.178      -5.59  2.28e- 8
## 3 Sexmale     -2.71      0.244     -11.1   1.40e-28
## 4 Age         -0.0415    0.00932    -4.46  8.24e- 6
## 5 SibSp       -0.407     0.142      -2.87  4.05e- 3
## 6 Parch       -0.0736    0.138      -0.534 5.93e- 1
## 7 Fare         0.00270   0.00262     1.03  3.02e- 1
```

Give an interpretation of the estimated slope of `Sex` in terms of the odds ratio.

c) Given the full model, we are now able to compute predictions for test dataset.

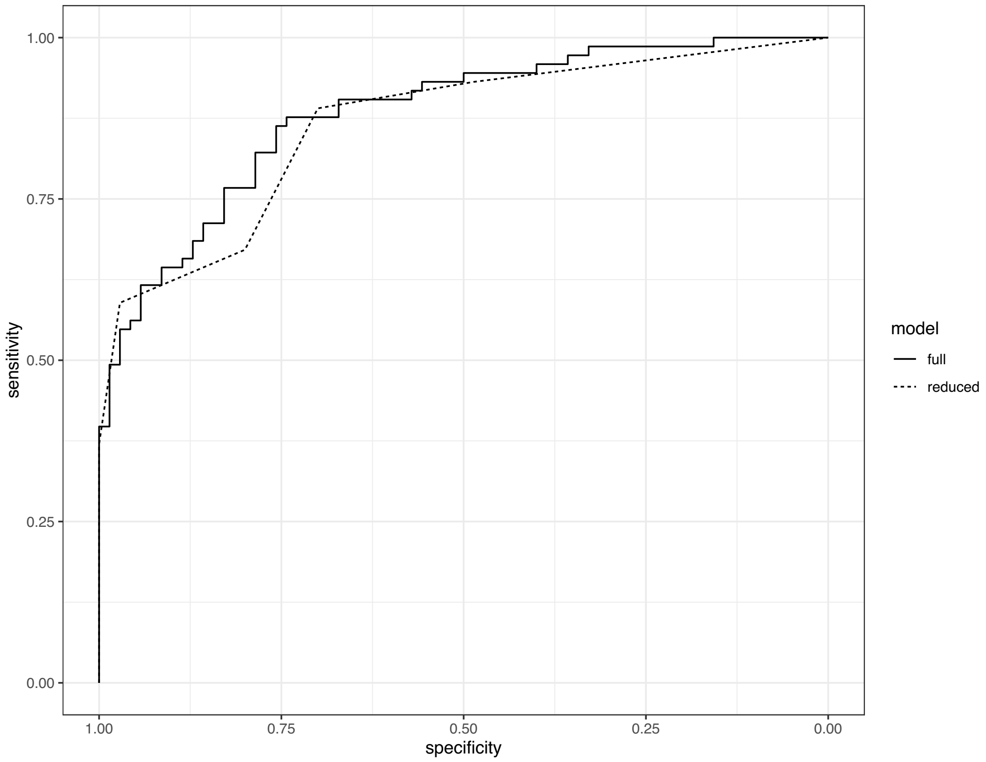
```
titanic_test$pred_prob <- predict(model_full, newdata = titanic_test, type = "response")
```

To predict survival we need a threshold for the computed probabilities of survival. Choosing a threshold of 0.5 leads to the following table

```
table(titanic_test$Survived, titanic_test$pred_prob > 0.5)
##
##      FALSE TRUE
##   0    58   12
##   1    18   55
```

Compute the sensitivity of this prediction.

d) Finally we compare the predictive accuracy of the model to a reduced model containing just the variables `Sex` and `Pclass`. The following graphic contains the ROC curves of the full and reduced model.



The area under the ROC curve of the full model is equal to 0.8818 and the reduced model leads to an area under the ROC curve of 0.8641. Which of the two models shows the better predictive accuracy with respect to the ROC curve? Justify your answer.

Cheat sheet

Summarizing data:

$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2 \qquad s = \sqrt{s^2} \qquad IQR = Q_3 - Q_1 \qquad Q_1 - 1.5 \cdot IQR, \quad Q_3 + 1.5 \cdot IQR$

Probability:

$P(A^c) = 1 - P(A)$; $\quad P(A \cap B) = P(A) \cdot P(B)$, $\quad A, B$ independent; $\quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$;

$E(X) = \sum_{i=1}^{k} x_i P(X = x_i)$, $\quad E(X) = \int_{\mathbb{R}} x \cdot f(x) \mathrm{d}x$; $\quad Var(X) = E(X^2) - E(X)^2$;

$SD(X) = \sqrt{Var(X)}$; $\quad Cov(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y)$; $\qquad Z = \frac{X - E(X)}{\sqrt{Var(X)}}$

Geom. dist.: $P(X = x) = (1-p)^{x-1}p$, $\quad x \in \mathbb{N}$; $\qquad E(X) = \frac{1}{p}$, $SD(X) = \sqrt{\frac{1-p}{p^2}}$

Binom. dist.: $P(X = x) = \frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}$, $\quad x \in \{0, 1, 2, ..., n\}$; $\qquad E(X) = np$, $SD(X) = \sqrt{np(1-p)}$

Neg. Binom. dist.: $P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!}p^k(1-p)^{x-k}$, $\quad x \in \mathbb{N}$;

Pois. dist.: $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$, $\quad x \in \mathbb{N}_0$; $\qquad E(X) = \lambda$, $SD(X) = \sqrt{\lambda}$

Inference for categorical data:

$\widehat{p} \pm z^* \cdot SE_{\widehat{p}} \qquad SE_{\widehat{p}} = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \qquad Z = \frac{\widehat{p} - p_0}{SE_0} \qquad SE_0 = \sqrt{\frac{p_0(1-p_0)}{n}}$

$(\widehat{p}_1 - \widehat{p}_2) \pm z^* \cdot SE \quad SE = \sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}} \quad Z = \frac{(\widehat{p}_1 - \widehat{p}_2) - 0}{SE_0} \quad SE_0 = \sqrt{\widehat{p}_{pooled}(1 - \widehat{p}_{pooled})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

$\widehat{p}_{pooled} = \frac{\sum x_{1,i} + \sum x_{2,i}}{n_1 + n_2}$; $\quad \chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$, $\quad$ expected $= \frac{\text{row} \times \text{column}}{\text{total}}$ (ind.), $\quad df = k - 1$ or $(r-1) \cdot (c-1)$

Inference for numerical data:

$\overline{x} \pm t^*_{df} \cdot SE_{\overline{x}} \qquad T_{df} = \frac{\overline{x} - \mu_0}{SE_{\overline{x}}} \qquad SE_{\overline{x}} = \frac{s}{\sqrt{n}} \qquad df = n - 1$

$(\overline{x}_1 - \overline{x}_2) \pm t^*_{df} \cdot SE \qquad T_{df} = \frac{(\overline{x}_1 - \overline{x}_2) - 0}{SE} \qquad SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \qquad df = \min(n_1 - 1, n_2 - 1)$

$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \qquad SE = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \qquad df = n_1 + n_2 - 2$

ANOVA and linear regression:

$F = \frac{MSG}{MSE} \qquad df_1 = k - 1$, $df_2 = n - k$, for k groups; $\qquad T_{df} = \frac{\overline{x}_i - \overline{x}_j}{\sqrt{\frac{MSE}{n_i} + \frac{MSE}{n_j}}} \qquad df = n - k \quad \alpha^* = \alpha / K$

$\widehat{y} = b_0 + b_1 x \qquad b_1 = R_{xy} \cdot \frac{s_y}{s_x} \qquad \overline{y} = b_0 + b_1 \overline{x} \qquad$ residual $= y - \widehat{y}$

$\widehat{\sigma} = \sqrt{\frac{1}{n-2}\sum(y_i - \widehat{y}_i)^2} = \sqrt{\frac{1}{n-2}\sum(y_i - b_0 - b_1 x_i)^2}$

$R_{xy} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{s_x \cdot s_y} \qquad T_{df} = \frac{b_1 - \beta_{1,0}}{SE_{b_1}} \qquad b_1 \pm t^*_{df} \cdot SE_{b_1} \qquad df = n - 2$

$R^2 = \frac{\sum_{i=1}^{n}(\widehat{y}_i - \overline{y})^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2} = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2}$; $R^2_{adj} = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2/(n-k-1)}{\sum_{i=1}^{n}(y_i - \overline{y})^2/(n-1)}$;

$AIC = \text{constant} + n \cdot \ln\left(\sum_{i=1}^{n} e_i^2\right) + 2k$

Logistic regression:

$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i} \qquad OR_j = \frac{\text{odds when } x_{i,j} = x+1}{\text{odds when } x_{i,j} = x} = e^{\beta_j}$

sensitivity = P(Test $+$ | Condition $+$), $\quad$ specificity = P(Test $-$ | Condition $-$)

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.